MIL-I-00705D (GL)

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MILITARY SPECIFICATION

ICE CREAM MIXES, REGULAR

AND IMITATION, DEHYDRATED

This limited coordination Military specification has been prepared by the U.S. Army Natick Laboratories, Natick, Mass., 01760, based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-I-705C. It is subject to modification. However, pending the promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE AND CLASSIFICATION

* 1.1 Scope.- This specification covers non-perishable ice cream and imitation ice cream mixes for use by the Armed Forces as items of general use (see 6.1).

* 1.2 Classification.-

Type I - Regular (containing milk fat)
Type II - Imitation (containing no milk fat)

Style A - Containing only coconut fat
Style B - Containing vegetable or animal fat or combinations thereof, other than coconut fat and milk fat

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-C-495 - Coatings, Exterior for Tinned Food Cans
PPP-B-636 - Box, Fiberboard
PPP-C-96 - Cans, Metal, 28 Gauge and Lighter
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

FSC 8910

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MILITARY

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit; Preparation of Nonperishable Subsistence In
MIL-F-35081 - Fat, Hydrogenated, Vegetable or Animal

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-671 - Sanitary Standards for Milk Evaporating and Drying Plants

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE

Federal Food, Drug, and Cosmetic Act As Amended and Regulations Promulgated Thereunder

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402).

U.S. DEPARTMENT OF AGRICULTURE

Visual Aids for Inspection of Metal Containers

Working Reference of Livestock Regulatory Establishments, Circuits, and Officials.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

U.S. Standard for Grades of Nonfat Dry Milk (Spray Process)

(Application for copies should be addressed to the Consumer and Marketing Service, Dairy Division, U.S. Department of Agriculture, Washington, D.C. 20250.)

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ASSOCIATION OF OFFICIAL AGRICULTURAL CHEMISTS

Methods of Analysis

(Application for copies should be addressed to the Association of Official Analytical Chemists, Box 540, Benjamin Franklin Station, Washington, D.C., 20044.)

AMERICAN OIL CHEMISTS' SOCIETY

Official and Tentative Methods

(Application for copies should be addressed to the American Oil Chemists' Society, 35 E. Wacker Drive, Chicago, Ill. 60601.)

AMERICAN PUBLIC HEALTH ASSOCIATION

Standard Methods for the Examination of Dairy Products

(Application for copies should be addressed to the American Public Health Association, 1790 Broadway, New York, N. Y. 10019.)

AMERICAN DRY MILK INSTITUTE, INC.

Standards for Grades of Dry Milks, Including Methods of Analysis Bulletin 916

(Application for copies should be addressed to the American Dry Milk Institute, Inc., 130 N. Franklin St., Chicago, Ill., 60606.)

MUNSELL COLOR CO., INC.

Munsell Color Discs

(Munsell Color Discs may be obtained from the Munsell Color Co., Inc., 2441 No. Calvert St., Baltimore, Maryland 21233.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Ill. 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

National Motor Freight Classification

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(Application for copies should be addressed to the American Trucking Association, ATTN: Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

3. REQUIREMENTS

* 3.1 First article (preproduction sample).- When specified (see 6.1), a sample of the finished product shall be submitted to the contracting officer for approval before production is commenced.

* 3.2 Material.-

* 3.2.1 Raw whole milk.- Raw milk shall be obtained from cows in herds accredited as tuberculosis-free and certified brucellosis-free by the U.S. Department of Agriculture (U.S.D.A.) or herds that have passed an annual tuberculosis test and meet U.S.D.A. requirements for an individually certified herd or from cows in herds located in (1) a Modified Accredited Tuberculosis Area and (2) either (a) a Certified Brucellosis-free Area, or (b) a Modified Certified Brucellosis Area; or (3) an area in the process of being accredited or certified by the U.S.D.A. In addition, the milk shall be normal in appearance; practically free from colostrum and have a clean sweet odor. It shall be subject to inspection by the procuring agency or duly authorized representative. The bacterial estimate at the time of processing shall not exceed 3,000,000 per milliliter (ml.) when determined by the standard plate count or direct microscopic clump count, or alternatively, methylene blue shall be decolorized in not less than 2 1/2 hours or resazurin reduced to Munsell color standard 5 p 7/4 in not less than 1 1/2 hours. The sediment content, when determined on a mixed sample shall not exceed 1.5 milligrams (mg.) per pint.

* 3.2.2 Skimmed milk, concentrated skimmed milk, and concentrated whole milk.- Skimmed milk, concentrated skimmed milk, and concentrated whole milk shall be prepared from raw whole milk conforming to the requirements of 3.2.1.

* 3.2.3 Cream, plastic cream, churned fat, anhydrous milk fat, and butteroil.- Cream, plastic cream, churned fat, anhydrous milk fat, and butteroil shall be prepared from raw whole milk conforming to the requirements of 3.2.1.

* 3.2.4 Nonfat dry milk.- Nonfat dry milk shall comply with the U.S. Standards for Grades of Nonfat Dry Milk (spray process) for Extra Grade, low heat powder.

* 3.2.5 Modified milk products.- Modified milk products shall be of edible grade derived by the treatment of milk solids to reduce lactose, calcium or minerals and shall be prepared from milk conforming to the requirements of 3.2.1.

* 3.2.6 Vegetable or animal fat.- Vegetable or animal fat other than coconut fat and milk fat shall conform to the requirements of MIL-F-35081, except that the Wiley melting point shall be $93^{\circ} \pm 3^{\circ}$F.

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* 3.2.7 Coconut fat.- Coconut fat shall comply with the following physical and chemical requirements.

* 3.2.7.1 Physical requirements.- Coconut fat shall have a clean bland flavor and odor free from extraneous materials. The liquified fat shall be clear and brilliant. The color shall be 15 yellow and 1.5 red maximum (Lovibond).

* 3.2.7.2 Chemical requirements.-

Moisture	None
Wiley melting point	93° ± 3°F.
Peroxide value, not higher than	1.0 meq/kg of fat
Free fatty acids (as oleic), not more than	0.05 percent
Iodine value, not higher than	9
Fat stability, (AOM) not less than	100 hours

* 3.2.8 Sweetening ingredients.- Sweetening ingredients, dry or liquid form, shall consist of sucrose, invert sugar, corn sugar, corn syrup solids used singly or in combinations.

* 3.2.9 Vanilla flavoring.- Vanilla flavoring shall be vanilla powder (sugar), vanillin or combinations thereof.

* 3.2.10 Stabilizers, stabilizing salts, emulsifiers, and coloring material.- Stabilizers, stabilizing salts, emulsifiers, and coloring material shall comply with 3.7.

* 3.2.11 Salt.- Salt shall be white, refined, noniodized sodium chloride.

* 3.3 Processing.-

* 3.3.1 Raw milk and dairy products.- All raw milk and dairy products not processed within two hours after receipt shall be cooled to 45°F. or lower and maintained thereat until processing is started.

* 3.3.2 Pasteurization.- The entire liquid basic mix, with the exception of part or all of the sugar and flavoring, shall be pasteurized by heating every particle to at least 155°F. and holding the mix at this temperature for at least 30 minutes or by any other pasteurization process recognized as equally efficient by the U.S. Public Health Service and approved by the appropriate State or local authority.

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* 3.3.3 Drying.- The liquid or concentrated mix shall be spray dried. The dried product shall be cooled to 100°F. or lower upon removal from the spray drier. Prior to packaging, the sugar and flavoring shall be blended into the product if these ingredients were not added to the liquid mix prior to pasteurization. Spray driers shall be of the continuous product removal type and wet collectors shall not be used.

* 3.3.4 Prepackaging holding requirements.- The finished product may be held for not more than seven days after manufacturing before final packaging. If not packaged in the final container within 24 hours after manufacture, the product shall be continually stored either in an atmosphere of inert gas or under a constant vacuum of not less than 20 inches. If the storage container is opened or the vacuum broken more than 24 hours prior to the final packaging, the above procedure shall be repeated. Dry blending of the product more than 24 hours prior to final packaging shall not be permitted unless the above operations are repeated. The storage containers shall not have surfaces of iron or copper exposed to the product.

* 3.4 Finished product.-

* 3.4.1 Physical requirements of dehydrated mix.-

* 3.4.1.1 Flavor and odor.- There shall be no foreign flavor or odor such as, but not limited to: oxidized, rancid, stale, tallowy, scorched or fishy.

* 3.4.1.2 Color.- The color shall be white or light yellow.

* 3.4.1.3 Foreign material.- There shall be no foreign material such as, but not limited to: dirt, insects or insect parts, wood, paint, glass or metal particles.

* 3.4.1.4 Lumps.- The product shall be free from lumps that do not fall apart under slight pressure.

* 3.4.2 Physical requirements of rehydrated mix and hardened ice cream.-

* 3.4.2.1 Reconstitution of the mix.- The mix shall reconstitute readily with potable water to a typical fluid ice cream mix.

* 3.4.2.2 Freezing performance.- The mix, when reconstituted in accordance with 5.4.1 and when frozen under suitable commercial conditions, shall have good whipping ability. Under these specified provisions, the mix shall yield a semi-frozen product having a stiff consistency and relatively dry appearance without affecting the attainment of 100% overrun in less than 12 minutes of freezing (batch method) at a drawing temperature of from

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21° - 24°F. The semi-frozen ice cream upon extrusion, shall be characteristically creamy and smooth in texture with a firm, compact body. Continuous frozen ice cream (when applicable) made from the mix shall result in a product of similar (or better) body and texture attributes as for batch frozen ice cream.

* 3.4.2.3 Hardened ice cream.- The hardened ice cream shall have a smooth texture and shall be free from palpable ice or lactose crystals. The flavor shall be pleasing, desirable and typical of vanilla ice cream.

* 3.4.3 Chemical requirements.- The product shall comply with the chemical requirements in Table I.

TABLE I.- Chemical Requirements

Item		Requirement
	Dehydrated mix	
1	Total fat, not less than	27.0 percent
2	Protein, not less than	9.75 percent
3	Moisture, not more than	2.0 percent
4	Oxygen in head space not more than	2.0 percent
5	Scorched particles not more than	15.0 milligrams
6	Iron 1/, not more than	10.0 parts per million (p.p.m.)
7	Copper 1/, not more than	1.0 p.p.m.

1/ If a copper and iron test made on the initial production lot complies with the specification requirement, subsequent testing shall be on a reduced basis. Reduced testing shall mean the testing of not less than one randomly selected sample every 6 months. If the initial production lot does not comply with the minimum copper and iron requirement, testing of subsequent lots in each contract shall be continued until two successive lots are in compliance with the requirements.

* 3.4.4 Microbiological requirements.- The dehydrated product shall comply with the microbiological requirements in Table II.

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TABLE II.- Microbiological requirements

Item		Requirement
1	Standard plate count, not more than	30,000 per gram
2	Coliform count, not more than	10 per gram
3	Yeast and mold (combined count) not more than	20 per gram
4	Salmonella test	Negative

* 3.5 Workmanship.- The product shall be prepared, processed, and packaged under modern sanitary conditions and in accordance with good commercial practice. The product shall be processed in establishments meeting the requirements of MIL-STD-671.

* 3.6 Delivery of product.- The product shall be delivered within 60 days after date of manufacture.

* 3.7 All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act As Amended and Regulations Promulgated Thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Pre-acceptance inspection.- Product shall be rejected if produced in plants not meeting the requirements of MIL-STD-671.

4.3 Sampling for inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105 except as otherwise indicated herein.

4.3.1 Component and material inspection.- In accordance with 4.1, components and materials shall be inspected in accordance with the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

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4.3.1.1 Inspection of milk and milk products.-

4.3.1.1.1 Animal sources.- Inspection shall be made to determine that cows supplying the milk used in the preparation of this product are in herds designated by the U.S. Department of Agriculture as tuberculosis-free and from herds designated brucellosis-free or in the process of being so designated. Determination of herd status regarding tuberculosis and brucellosis shall be made by contacting the nearest Office of the Animal Agricultural Research Service of the Animal Health Division. (Location of this office can be ascertained by consulting the publication entitled "Directory of Consumer Protection Programs, Establishments, Circuits, and Officials".) Nonconformance to either of the above requirements shall be cause for rejection of the lot.

4.3.1.1.2 Establishment sources.- Examination shall be made to determine that all processed milk products have been manufactured, packaged and shipped by establishments approved as Military sources of supply as listed in the Directory of Sanitarily Approved Food Establishments for Armed Forces Procurement, or from plants listed with a rating of 90 or more in the document entitled "Sanitation Compliance and Enforcement Ratings of Interstate Milk Shippers". Nonconformance to the above requirements shall be cause for rejection of the lot or the involved quantity of finished product made therefrom, as applicable.

4.3.1.2 Examination of raw whole milk.- Examination and testing of raw whole milk shall be made to determine compliance with the requirements of 3.2.1 as concerns appearance and odor, bacterial estimate and sediment at the approximate time of processing. Records of all required examinations and tests shall be maintained and made available for review. Failure to comply with one or more of the above referenced requirement(s), reflected by actual examination and tests or by examination of records, shall be cause for rejection of the involved lot of finished product.

4.3.1.3 Examination of milk and modified milk product ingredients.- Conformance of dairy product ingredients to the requirements of 3.2.2 through 3.2.5 as concerns identity (kind, raw milk supply or applicable grade) shall be determined by examination of pertinent labels, invoices, USDA grade certificate, plant records reflecting appearance, odor, bacterial estimate and sediment data on original raw milk, or other valid documents. In addition, these products shall be examined organoleptically, as necessary, to determine conformance to the above referenced condition requirements. The sample for examination shall be approximately 8 ounces (fluid or avoirdupois, as applicable) of ingredient taken from each of 5 primary containers or all primary containers if less than 5 form a lot. Failure to conform to one or more of the above referenced requirements, reflected (as applicable) by actual examination and testing or by examination of records, or to one or more identity or condition requirements, shall indicate an unacceptable ingredient(s) and use of such shall be cause for rejection of the involved end item.

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4.3.1.4 Examination of vegetable or animal fat.- Examination of vegetable or animal fat shall be made in accordance with the examination criteria for product characteristics of MIL-F-35081, except that the sampling plans and acceptable quality levels (AQLs) shall not apply. The sample for examination shall be approximately 1-pound composite derived from 5 primary containers or all containers if less than 5 form a lot. Nonconformance to one or more requirements shall be cause for rejection of the involved lot of finished product.

4.3.1.5 Examination of coconut fat.- Conformance of this fat to the requirements of 3.2.7 as concerns identity (kind and type) shall be determined by examination of pertinent labels, invoices or other valid documents. In addition, each fat shall be examined organoleptically, as necessary, to determine conformance to the condition requirements as concerns flavor, odor, appearance, and presence of extraneous material. The sample for examination shall be approximately 8 ounces taken from each of 5 primary containers or all containers if the lot contain less than 5 containers. Nonconformance to one or more identity or condition requirements shall indicate an unacceptable fat(s) and use of such shall be cause for rejection of the involved lot of end item.

4.3.1.6 Examination of sweeteners, vanilla flavoring, stabilizers, stabilizing salts, emulsifiers, salt and coloring material.- Examination of these products shall be made to determine conformance to the requirements of 3.2.8 through 3.2.11, as concerns identity (kind and types). This determination shall be ascertained by examination of labels, invoices or other valid documents. In addition these products shall be examined organoleptically, as necessary, to determine conformance to the condition characteristics in accordance with the pertinent paragraph requirements. The sample for examination shall be approximately 8 ounces of ingredient except dry flavor and color, when used, for which the sample shall be 1-2 ounces, taken from each of 5 primary containers or all containers if the lot contains less than 5 containers. Nonconformance to one or more identity or condition requirements shall indicate an unacceptable ingredient(s) and use of such shall be cause for rejection of the involved lot of end item.

4.3.2 Testing of components.- Testing of components described in this specification shall be as shown in table III. Results shall be applicable to the lot average. Nonconformance to one or more test requirements shall be cuase for rejection of the lot.

TABLE III.- Testing of components 1/

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported	Test Ref & reqmt para
Cans & lids 2/	1 can and 1 lid	Cans	S-1	Tinplate	Nearest 0.01 lb/ base box	4.5.2.4 and 5.1.1

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TABLE III.- Testing of components 1/ (cont'd)

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported	Test ref & reqmt para
Coconut fat	3/	Primary containers	3/	Color	As less than minimum or greater than maximum	4.5 and 3.2.7.1
				Moisture	Pass or fail	4.5 and 3.2.7.2
				Wiley melting point	Nearest degree Fahrenheit	
				Peroxide value	Nearest 0.1 meq./kg of fat	
				Free fatty acids (as oleic acid)	Nearest 0.01 percent	
				Iodine value	Nearest whole number	
				Fat stability (A.O.M.)	Pass or fail 4/	

1/ Each component or component characteristic, as applicable, which is not listed in this table, unless otherwise accepted, shall be tested in accordance with the applicable procedure contained in pertinent subsidiary specifications.

2/ Exterior can coating, when required, and can label shall be tested in accordance with the applicable subsidiary specifications except that tests shall be performed on the same cans submitted for tinplate testing.

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3/ The sample for testing shall be an approximate 1-pound composite derived from 5 primary containers or all containers if less than 5 form a lot.

4/ If failure, report to nearest hour.

4.3.3 Process examination.- Examination shall be made during processing to determine compliance with requirements of 3.3.1 through 3.3.3 as concerns time, time intervals, temperature, equipment, procedures and controls. Time and temperature records shall be maintained. Non compliance with the above requirements, reflected either by examination of records or actual examination, shall be cause for rejection of the involved quantity of finished product.

4.3.3.1 Examination of pre-packing requirements.- Examination shall be made to determine compliance with pre-packing requirements of 3.3.4 as concerns storage conditions (use of inert gas or vacuum), storage time before final packaging, vacuum control, dry blending time limitations (when applicable) and type of storage containers. Storage time and vacuum records shall be maintained. Noncompliance with the above referenced requirements as reflected by examination of records or by actual examination shall be cause for rejection of the lot.

4.3.4 Examination of end item.- Examination of end item shall be performed in accordance with tables V through VIII and 4.3.4.1 through 4.3.4.5. The examination criteria for the above mentioned tables shall be as shown in table IV and the acceptable quality levels (AQLs) for tables V and VIII shall be expressed as defects per hundred units and for tables VI and VII, as percent defective.

TABLE IV.- End item examination criteria

Table	Inspection level	Sample unit	Lot size expressed in	AQLs		
				Major	Minor	Total
V	I	1 filled and sealed can	Cans	1.5	-	4.0
VI	S-3	1 filled and sealed can	Cans	-	2.5	-
VII	S-3	1 filled and sealed can	Cans	0.65	-	-
VIII	S-2	Contents of 1 filled can	Cans	2.5	-	-

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TABLE V.- External examination of cans 1/

Category		Defect
Major	Minor	
101		Hole or leaker.
102		Improperly closed.
103		Not type, size, or style specified.
104		Severe dent.
105		Dent causing sharp ridge.
	201	Moderate dent.
	202	Not clean. 2/
106		Cable cut.
107		Pitted rust. 3/
	203	Excessive amount of flux on ends or sides. 4/

1/ Use USDA-CMS Visual Aids for Inspection of Metal Containers as a guide for classifying can defects.

2/ Cans showing a very thin film of grease which is discernible to the touch but not readily discernible visually are considered to be clean.

3/ Rust that can be removed with a soft cloth will not be considered as pitted rust.

4/ Traces of flux at edges of soldered areas normally encountered in can manufacture but not enough to cause corrosion will not be considered excessive.

TABLE VI.- Examination for net weight 1/

Category	Defect
Minor	
201	Net weight less than 4 lbs., 2.5 ounces 2/

1/ Lot shall be rejected if sample data indicate lot average net weight is less than the specified net weight.

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2/ Report results to the nearest 1/4 ounce.

TABLE VII.- Examination for can leakage (see 4.5.1.2)

Category	Defect
Major	
101	Steady progression of bubbles.

TABLE VIII.- Examination of product characteristics

Category	Defect
Major	Dehydrated mix 1/
101	Color not white or light yellow.
102	Type and style not as specified.
103	Not uniform in appearance or not well blended.
104	Presence of lumps which do not fall apart under light pressure.
	Rehydrated mix 2/
105	Does not reconstitute readily to a typical fluid ice cream mix.
	Freezing performance (see 3.4.2.2) 3/ 4/
106	Semi-frozen product drawn from freezer lacks a stiff consistency and relatively dry appearance.
107	Extruded ice cream not characteristically creamy and smooth in texture or lacking in a firm, compact body.
108	Overrun attainment (batch frozen) not as specified.
	Hardened ice cream 4/
109	Lacks smooth texture or not free from palpable ice or lactose crystals (sandiness).
110	Flavor not pleasing or characteristic of vanilla ice cream.

1/ Presence of foreign flavor or odor (e.g. oxidized, rancid, stale, tallowy, scorched, fishy, etc.) or presence of foreign material, (e.g. dirt, insect(s), insect part(s), wood, paint, glass, metal, etc.,) shall be cause for rejection of the entire lot.

2/ See 3.4.2.1 and 5.4.1 for reconstitution instructions.

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3/ For purposes of freezing reconstituted powder, select a minimum of 3 primary containers from each lot, disregarding sample size indicated by inspection level S-2 for table VIII cited in table IV.

4/ Sampling and examination for these requirements shall be performed jointly by the Government and the supplier during Government inspection.

4.3.4.1 Examination of exterior can coating (when applicable).- Examination of exterior can coating shall be performed in accordance with the applicable examination criteria of TT-C-495.

4.3.4.2 Examination of can labeling.- Examination of can labeling shall be in accordance with the examination criteria of MIL-L-1497.

4.3.4.3 Examination of age of product at time of delivery.- Examination shall be made at time of delivery to determine compliance with 3.6. This determination shall be ascertained by examination of records, codes, marked date of pack or other valid means. Noncompliance with the above referenced requirement shall be cause for rejection of the lot.

4.3.4.4 Examination of shipping container.- Examination of the filled and closed shipping container for levels A and B packing shall be made in accordance with the examination criteria in the appendix to PPP-B-636. In addition, the following defects shall be included in the table of examination: Major - markings missing, incorrect or illegible; Minor - arrangement not as specified. Also, for level B, containers with round wire strapping intended for transfer at sea shall not be acceptable. For level C packing, examination shall be in accordance with PPP-B-636, except that only the defects for marking and arrangement shall apply.

4.3.4.5 Examination of palletized loads.- Palletized loads shall be examined in accordance with the criteria of MIL-L-35078.

4.3.5 Sampling procedures and acceptance criteria for testing of finished product.- The finished product (dehydrated mix) shall be tested for total fat, protein, moisture, scorched particles, iron, copper (see footnote 1/ to table I), oxygen and microbiological content in accordance with the requirements of 3.4.3, 3.4.4 and 5.1.1. Procedures for testing shall be in accordance with 4.5. Test requirements for oxygen and microbiological content shall be applicable to the sample unit. The sample unit for testing shall be one filled and sealed primary container. Lot size shall be expressed in terms of primary containers and the sample size shall be the number of sample units indicated by inspection level S-1. The AQL, expressed as percent defective, and applicable to the oxygen test only shall be 1.5. A defect shall be scored if a test result exceeds 2.0 percent oxygen. The sample unit for testing for fat, protein, moisture, scorched particles, iron and copper, shall be approximately a 1-pound composite derived from the samples used for unit testing. Results shall be reported to the same unit or decimal as specified for the pertinent requirements except that all microbiological counts shall be reported in accordance with Standard Methods for Examination of Dairy Products and the presence of Salmonellae as positive or negative. Nonconformance to one or more of the above requirements, as applicable, shall be cause for rejection of the lot.

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4.4 Standby test samples.- The Government reserves the right to withdraw and hold standby test samples of components or finished product or both (quantity of which shall be not more than twice that required by this specification) for inspection purposes. Samples not used will be returned to the supplier.

4.5 Test methods.- Examination and test procedures which differ from those specified herein, unless otherwise excepted, may be used by the supplier if they provide a quality assurance equivalent to that specified. If the Government inspection activity determines that such procedures and controls do not provide, as a minimum such quality assurance, the supplier will use the procedures set forth herein. In case of dispute as to examination or test results, the procedures specified herein will govern.

4.5.1 Examination procedures.-

4.5.1.1 Freezing performance examination.- The freezing performance of the mix shall be conducted as follows:

(a) Reconstitute the ice cream mix and chill to 36-38°F.
(b) Freeze the chilled mix in an ice cream freezer.
(c) Remove the frozen product into a container and place immediately into a hardening cabinet which has a temperature of less than 0°F. Allow to harden for 24 hours.
(d) Overrun shall be computed by using the following formula:

$$\frac{A - B}{B} \times 100 = \text{percent overrun}$$

A = weight of unit volume of ice cream mix
B = weight of same volume of frozen ice cream

4.5.1.2 Leakage.- The filled and sealed container shall be examined for leakage by submerging in water, contained in a desiccator or other suitable container and maintaining a vacuum of 10 inches of mercury (atmospheric pressure 29.9 inches) for at least 30 seconds. A leak is indicated by a steady progression of bubbles. Isolated bubbles caused by entrapped air not considered as leakers.

4.5.2 Test procedures.-

4.5.2.1 Chemical analyses.- Chemical analyses shall be made in accordance with the methods and publications listed below:

Test	Source	Method
Ingredients:		
Peroxide value	1/	Cd 8-53
Free fatty acids	1/	Ca 5a-40
Moisture	1/	Ca 2e-55

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Test	Source	Method
Ingredients:		
Iodine value	1/	Cd 1-25
Wiley melting point	1/	Cc 2-38
Fat stability (AOM)	1/	Cd 12-57
Color (coconut fat)	1/	Cc 13b-45
Monoglyceride content	1/	Cd 11-57
Finished product:		
Fat	2/	Fat in dried milk
Protein	2/	Protein
Moisture	3/	Toluene distillation
Iron	4/	Iron
Copper	3/	Copper
Scorched particles	3/	Scorched particles (22g. sample)
Oxygen in headspace	3/	Oxygen

1/ Official and Tentative Methods of the American Oil Chemists Society, Section C, Commercial Fats and Oils.

2/ Official Methods of Analysis of the Association of Official Agricultural Chemists (A.O.A.C.). Chapter: Dairy Products; Section: Dried Milk and Malted Milk.

3/ American Dry Milk Institute Bulletin 916, Standards for Grades of Dry Milks, including Methods of Analysis.

4/ A.O.A.C. - Chapter: Cereal Foods; Section: Wheat Flour.

* 4.5.2.2 Microbiological analyses.- Microbiological analyses shall be performed in accordance with the following methods from Standard Methods for the Examination of Dairy Products except as noted:

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Test	Chapter	Method
Ingredients:		
Standard Plate Count	Agar Plate Method	Agar Plate
Direct Microscopic	Direct Microscopic	Direct Microscopic
Clump Count	Method	Clump Count
Methylene Blue Reduction	Reduction Methods	Methylene Blue Reduction Method
Resazurin Reduction	Reduction Methods	Resazurin Reduction Method
Coliform Count	Coliform Bacteria	Coliform Bacteria (Desoxycholate lactose agar)
Yeast and Mold Count	Microbiological Methods for Butter	Yeast and Mold count
Finished Product:		
Standard plate count	Microbiological Methods for Concentrated Milk and Dry Milk	Agar Plate
Coliform Count	Loc. cit.	Coliform Group (Desoxycholate lactose agar)
Yeast and Mold Count	Loc. cit.	Yeast and Mold count
Salmonella test	1/	

1/ U.S. Department of Health, Education and Welfare, Bacteriological Analytical Manual, Section 14.7.

4.5.2.3 Sediment analyses.- Sediment content of the raw milk shall be in accordance with the procedures in Standard Methods for the Examination of Dairy Products; chapter: Sediment in Fluid Milk. Milk in cans shall be tested by the off-the-bottom method. For bulk milk, a mixed one-pint sample shall be used.

* 4.5.2.4 Tin coating weights.- Tin coating weights shall be determined by any method specified in PPP-C-96.

* 5. PREPARATION FOR DELIVERY

* 5.1 Packaging.- The product shall be packaged in accordance with level A or C, as specified (see 6.1).

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* 5.1.1 Level A.- Four pounds and 4 ounces of the type and style of ice cream mix specified (see 6.1), shall be filled into a size 603 by 700, open-top style, round, metal can with soldered side seam and compound-lined, double-seamed end. The can shall be made throughout from not less than commercial 0.25-pound electrolytic tin plate per base box (see 4.5.2.4) and shall be coated outside with a coating conforming to type I or, when specified (see 6.1), type III of TT-C-495. The ice cream mix shall be gas packaged by first removing the air from the filled can and replacing it with nitrogen. If tip soldering is used to close the exhaust vents for nitrogen filling, a suitable noncorrosive flux shall be used. When the end of the can is clinched prior to vacuumizing and gassing, the flange end of the can or the lid shall be indented or dimpled to prevent seal-off trouble. The can shall be hermetically sealed and shall not leak when examined in accordance with 4.5.1.2. The oxygen content of the gases at whatever pressure is in the filled and sealed container, not sooner than 2 days nor more than 14 days after filling, shall not exceed 2.0 percent when tested in accordance with 4.5.2.1.

* 5.1.2 Level C.- The ice cream mix shall be packaged in accordance with 5.1.1, except that cans with or without commercial exterior coating will be acceptable.

* 5.2 Packing.- Six cans of the ice cream mix, arranged 3 in length, 2 in width, and 1 in depth, shall be packed on end in a snug-fitting shipping container in accordance with level A, B, or C, as specified (see 6.1). RSC containers may be top-opening or end-opening.

* 5.2.1 Level A.- The shipping container shall be a fiberboard box, constructed and closed in accordance with style RSC, V2s, of PPP-B-636.

* 5.2.2 Level B.- The shipping container shall be a fiberboard box, constructed and closed in accordance with style RSC, type CF or SF, class domestic, method II closure, of PPP-B-636.

* 5.2.2.1 When specified (see 6.1 and 6.3), the shipping container shall be a fiberboard box, constructed and closed in accordance with style RSC, V3c or V3s, of PPP-B-636. Alternatively, the shipping container may be constructed from fiberboard complying with V4s of PPP-F-320. For transfer at sea, the shipping container shall be reinforced with other than round wire strapping in accordance with the appendix of PPP-B-636.

* 5.2.3 Level C.- The cans of ice cream mix shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The shipping container shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Rules, as applicable.

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* 5.3 Unit loads.- The ice cream mix, packed as specified in 5.2. shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1).

* 5.4 Labeling and marking.-

* 5.4.1 Cans.- Cans shall be labeled in accordance with MIL-L-1497. In addition, the following directions for use and explanatory statements shall be included:

> Directions: Pour contents of can (4-1/4 lb.) on 3-1/2 quarts of water, mix well, and freeze. Never add warm (over 100°F.) mix to freezer.
>
> Caution: Do not overheat or heat directly on stove or open flame. Never add warm (over 100°F.) mix to freezer.
>
> NOTE: This product is gas packed.

* 5.4.2 Shipping containers.- Shipping containers shall be marked in accordance with MIL-STD-129.

* 5.4.3 Unit loads.- Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

* 6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and style of ice cream mix required (see 1.2 and 5.1.1).
(c) First article (preproduction sample).- When a preproduction sample is required, (see 3.1).
(d) When type III exterior can coating is required, (see 5.1.1).
(e) Levels of packaging and packing required (see 5.1 and 5.2).
(f) When packing specified in 5.2.2.1 is required.
(g) Type and class of unit load required (see 5.3).

* 6.2 Product information.- Although refrigeration is not necessary, stability is improved if product is stored in a cool location. Type II product is a dehydrated mix used to make a frozen dessert identical to ice cream except that it contains edible fats of a non-dairy origin in lieu of milk fat. Type II product is intended as a substitute for type I (containing milk fat), when authorized, in areas where its manufacture and sale is not prohibited by law.

6.3 Packing specified in 5.2.2.1 is intended to provide limited Military protection for specific operational requirements.

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* 6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 45

Preparing activity:

Army - GL

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